

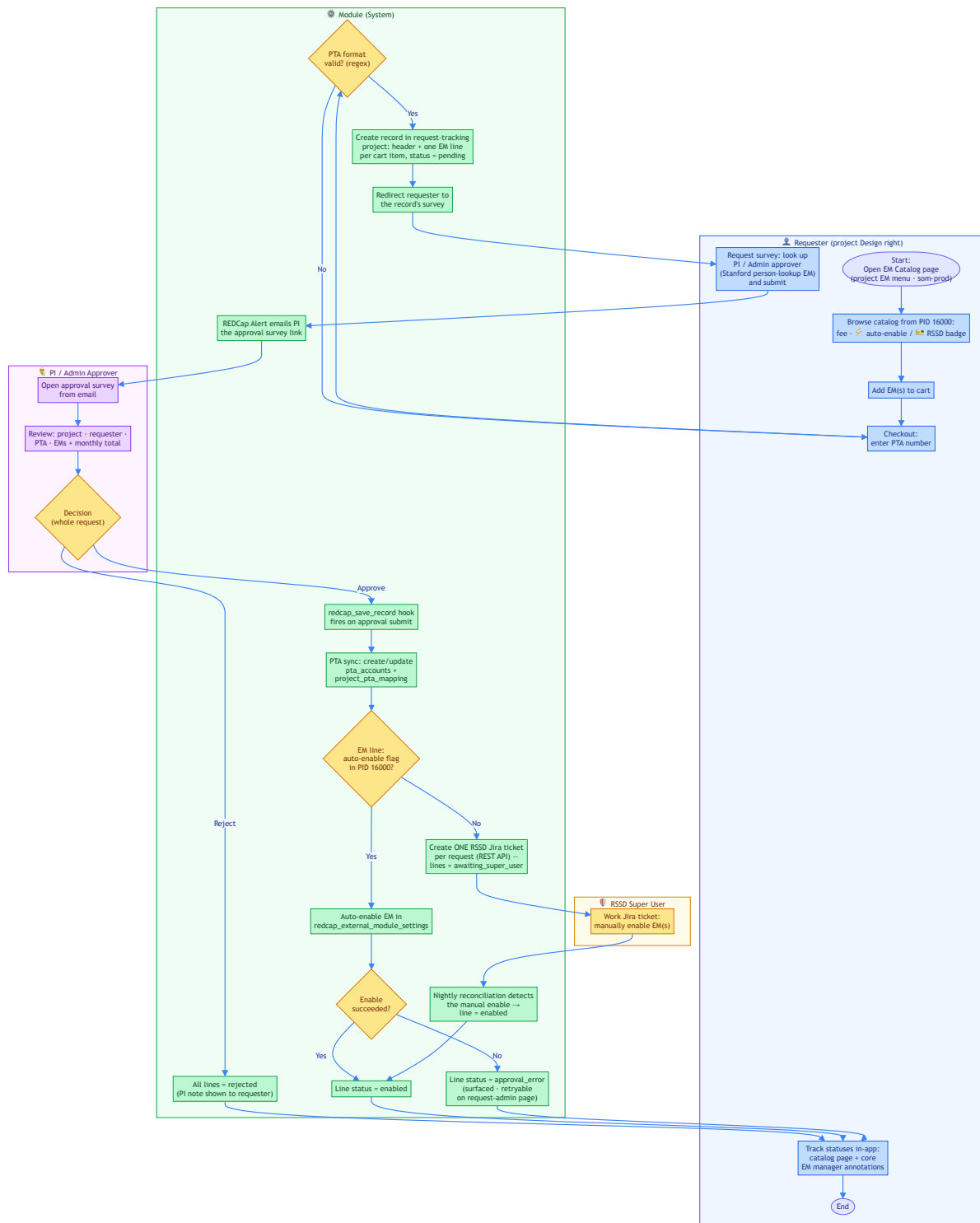
External Module Manager — Workflow Diagrams

v2.0 · Generated June 29, 2026 · updated July 6, 2026 · **revised July 10, 2026** (request flow redesigned: cart → survey → PI approval; per-PTA finance CSV; PTA debt ledger) · stanford/external_module_manager

Diagram 1 — EM Enablement Request: Cart → Survey → PI Approval

Redesigned July 10, 2026 — replaces the super-user approval queue. A requester with the project's "Design" right (som-prod, v1) opens the EM Catalog page in the project's EM menu, adds fee-bearing EMs to a cart (catalog + fees + auto-enable flag come from the EM-records project, PID 16000), and checks out with a regex-validated PTA. The module pre-creates a record in the *request-tracking REDCap project* and redirects to its survey, where the requester picks the PI/Admin approver via the Stanford person-lookup EM. A REDCap Alert emails the PI an approval survey — **one**

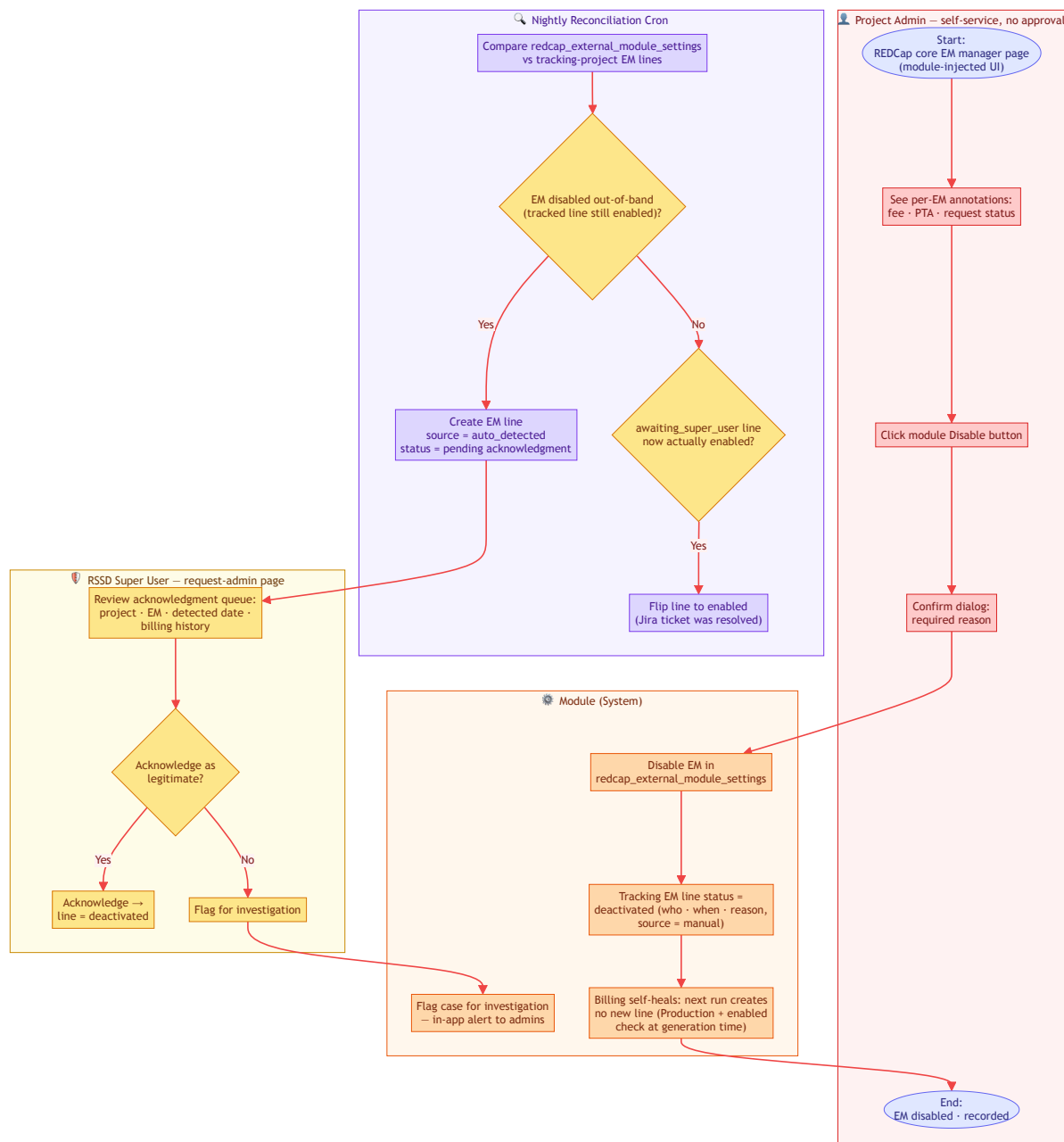
Approve/Reject decision for the whole request. On approval, the `redcap_save_record` hook syncs the PTA into the billing mapping, auto-enables flagged EMs (failure → `approval_error`, retryable), and opens **one RSSD Jira ticket per request** for the rest (`awaiting_super_user` until nightly reconciliation sees the manual enable). Re-enabling a deactivated EM (Diagram 2) comes back through this same flow with no shortcut.



- Requester action
- System / automated action
- PI / Admin approver (survey via email)
- RSSD super user (Jira) / decision
- Start / End terminal

Diagram 2 — EM Deactivation & Nightly Reconciliation

Revised July 10, 2026. Deactivation is self-service and immediate — via the module's own *Disable* button injected into REDCap's **core EM manager page** (each EM row is annotated with fee · PTA · request status). The confirm dialog requires a reason; the module disables the EM and records the deactivation on the tracking record (who · when · reason). No approval gate, no billing step: the monthly run only creates a line when the project is Production *and* the EM is enabled, so a deactivated EM stops generating new lines on its own — already-open lines run their normal course. The nightly reconciliation cron does two jobs: it files out-of-band disables as `auto_detected` records for super-user acknowledgment, and it flips `awaiting_super_user` lines to `enabled` once the Jira-ticketed manual enable actually happened. At go-live, a one-time seeding creates *grandfathered* tracking records for every currently enabled billable EM so reconciliation starts from a complete baseline. Re-enabling requires the full Diagram 1 flow again.



💰 Diagram 3 — Monthly Billing Run, Finance CSV Loop & Debt Ledger

Revised July 10, 2026. Generation is admin-triggered (day 28/29 is guidance, not a gate). One run per month: finalized/exported blocks regeneration, a draft

- RSSD super-user action
 ■ System / automated action
 ■ Finance team action (offline)
- Outstanding Balances ledger / backcharge
 ■ Decision / gate
 ■ Start / End terminal

CSV Contract — Two Files (revised July 10, 2026)

Finance receives **two files**. They edit and return only the **Summary CSV** — one row per PTA, because Finance charges/rejects whole PTAs. On import, each PTA row's status fans out to all of that PTA's still-pending lines in the run. The **Detail CSV** (one row per project+EM line) is a read-only reference and is never re-imported. Amounts are read-only everywhere (Migration Plan §5, decision #9).

File 1 — Summary CSV (editable, re-imported; one row per PTA)

#	Column	Direction	Description	Valid values
1	billing_run_id	Export only	Month's billing run entity ID	integer
2	billing_month	Export only	Month charged (YYYY-MM)	2026-07
3	pta_number	Export only — re-import key	PTA to charge (one row per PTA)	string
4	pta_description	Export only	PTA account description	string
5	contact_name	Export only	PTA contact	string
6	project_count	Export only	Projects billed under this PTA	integer
7	line_count	Export only	EM lines under this PTA	integer
8	new_amount	Export only	This month's new charges (USD)	integer
9	carried_amount	Export only	Carry-forward from prior months (USD)	integer
10	total_amount	Export only	new + carried (USD)	integer

#	Column	Direction	Description	Valid values
11	status	 Finance fills in	Outcome for this PTA — fans out to all its pending lines	charged · rejected · waived · hold
12	finance_note	 Finance fills in	Reason (required if rejected)	free text

File 2 — Detail CSV (read-only reference; one row per project+EM line; never re-imported)

line_id · pta_number · project_id · project_title · module_prefix · amount · attempt_number · carried_from_line_id · status

● Highlighted rows = columns Finance must fill in the Summary CSV before returning it. All other columns are read-only (pta_number is the re-import key). Unmapped-project lines (status hold, no PTA) are excluded from the Summary CSV and flagged at export time.

Entity & Status State Reference (revised July 10, 2026)

Request-tracking project — EM line status flow (REDCap records, not entities)

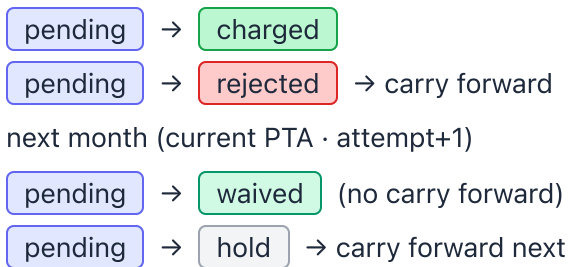
pending → *PI approves whole request* → approved
 approved → enabled (*auto-enable flag set in PID 16000*)
 approved → awaiting_super_user
 (RSSD Jira) → enabled
 approved → approval_error (*enable failed — surfaced, retryable*)
 pending → rejected (*PI rejects — new request to retry*)
 enabled → deactivated (*self-service or acknowledged auto-detect*)

Deactivation sources

source = manual — module Disable button on REDCap's core EM manager page; recorded already-resolved as deactivated, no queue entry (decision #13)
 source = auto_detected — nightly reconciliation finds an out-of-band disable:
 pending ack → deactivated (*super-user acknowledges, decision #14*)
 source = grandfathered — one-time go-live seeding of already-enabled billable EMs, created as enabled

finance_billing_lines status flow

finance_billing_runs status flow



*Any status may also be set manually by a
 super user (required note; status_source
 = manual; allowed post-finalize).*



One run per month. Draft regenerates in
 place; exported/finalized blocks regeneration
 (decision #12).

Data stores (revised)

- Request-tracking REDCap project** —
 request header + approval instruments +
 repeating EM-line instrument (fee/auto-
 enable snapshots, line status, Jira key,
 source). ~~em_enable_requests/~~
~~em_enable_request_items~~ entities —
 superseded, not built.
- finance_billing_runs** — month/year,
 status, totals,
 generated/exported/imported stamps.
- finance_billing_lines** — run_id,
 project, module, amount, snapshot PTA,
 status, status_source (generation |
 csv_import | manual) + who/when,
 attempt_number,
 carried_from_line_id, charge-entity
 link.
- ~~monthly_pta_reports~~ — replaced by
 runs + lines (JSON snapshot cannot track
 per-line status).
- Reuse:* pta_accounts,
 project_pta_mapping (kept up to date
 by approval-time PTA sync).

- *Extend:* `external_modules_charges` —
add source (live | migrated_r2p2),
`r2p2_charge_item_id`,
`r2p2_invoice_item_id` for the
historical migration (Migration Plan §8).
- *Catalog:* EM-records project (PID 16000)
— existing fee fields + new `auto_enable`
/ `requestable` flags.